

Security Protocols Tasks

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Schedule

1 Day 1

- Initial setup
- Traffic Analysis
- IPSEC
- JWT

2 Day 2

- PGP
- TLS
- 2FA
- OAuth 2.0

Deliverables

- 1 Solve as many tasks as you can
- 2 Document everything with screenshots in a doc/docx file
- 3 Convert it into a PDF
- 4 Sign the obtained PDF
- 5 Rename it to
HW_**SSP**_2024_'FamilyName'_'GivenName'.pdf
(ex: HW_**SSP**_2024_Popa_Ana.pdf)
- 6 Send it to teme.aciobanitei.iulian@gmail.com
Email Subject: HW_**SSP**_2024_'FamilyName'_'GivenName'

Initial setup

- 1 Install [Virtual Box](#)
- 2 Download and import the [Kali-based customized VM](#)

Traffic Analysis

- 1 Download and unzip the [pcap](#). Password is "infected".
- 2 Walk through [Changing your column display](#)
- 3 Walk through [Identify hosts and users](#)
- 4 Walk through [Display Filter Expression](#)
- 5 Walk through [Exporting objects from a pcap](#)
- 6 More details are [here](#).

Produce a document with your findings. It should contain:

- 1 **Executive summary.** What happened (who, when what)
- 2 **Victim details** - hostname, IP address, MAC address, Windows user account name.
- 3 **IOCs** - IP addresses, domains, URLs, files, etc.
- 4 **Screenshots** with your findings.

Network Miner

- 1 Install [Network Miner](#).
- 2 Download [pcap](#). Password is "infected".
- 3 Analyse the same pcap using Network Miner

IPSEC - initial setup

- 1 Use Kali VM already created
- 2 Setup the vm in bridge mode
- 3 Choose a partner
- 4 Ping to your partner's IP address
- 5 Change hosts file and name partner's IP
- 6 Start Wireshark
- 7 Monitor (ICMP) traffic from host to host generated by ping cmd

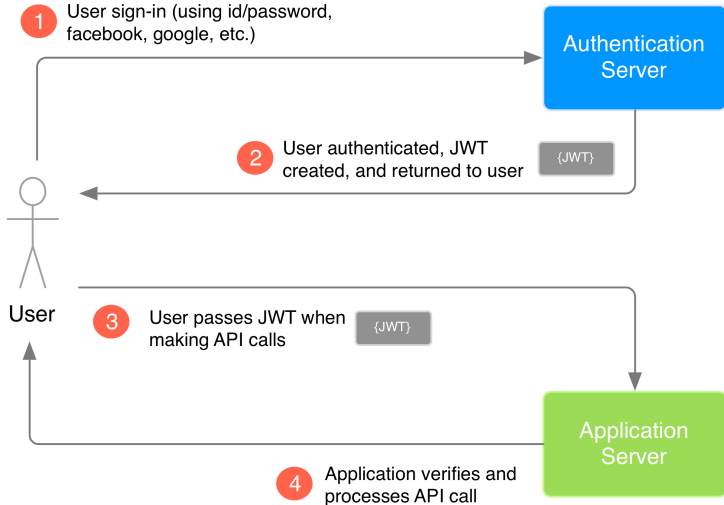
IPSEC documentation

- Check [ipsec.conf file documentation](#)
- Check [host-to-host](#) configuration example
- Install strongswan (already installed)
- Create AH tunnel based on PSK (PreShared Key)
- Create ESP tunnel based on PSK
- Create ESP tunnel based on certificates
- Check [how to generate a certificate guide](#)
- Files location:
 - /etc/ipsec.conf
 - /etc/ipsec.secrets
- Useful cmds:
 - systemctl status ipsec
 - systemctl restart ipsec

Secure HTTP traffic

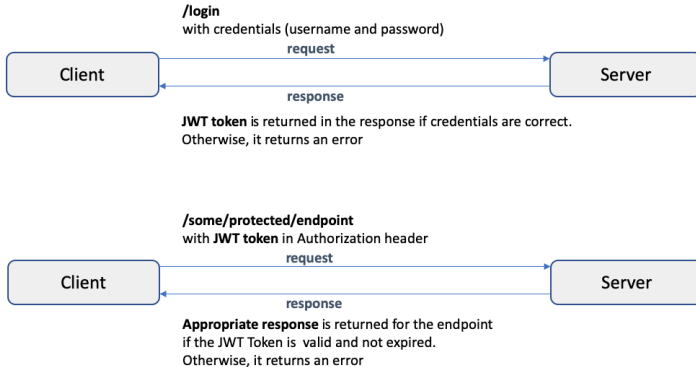
- Stop ipsec service
- Install Apache server
- Create a new html web page
- Start Apache server
- Observe clear HTTP traffic and webpage content
- Secure communication using IPSEC

Overview

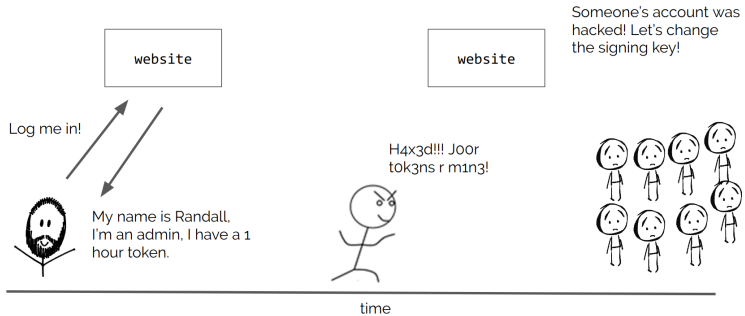


Resource Protection

JSON Web Token Authentication Overview



Key renewal

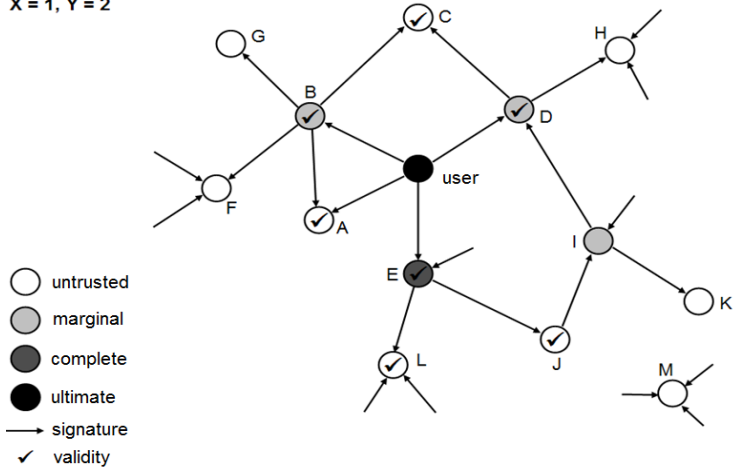


JWT - To do

- Install&Configure [BurpSuite](#)
- Start docker & webgoat
 - in Desktop folder, run 'sudo start_webgoat.sh'
- Create account on: <http://localhost:8080/WebGoat>
- Go to lesson (A2) Broken Authentication -> JWT Tokens
- At minimum, complete challenges 4 & 5
- At best, complete also 7 & 8

Web of trust

X = 1, Y = 2



PGP Demo

- PGP App Installation (`sudo apt-get install Kleopatra`)
- Key & cert generation
- Sign
- Encrypt
- Trust Transfer

PGP - To do

- Install PGP App
- Generate Key & cert
- Sign
- Encrypt
- Exchange certs to create the Web Of Trust
- Validate a Signature

TLS

- Demo IIS
- Demo apache2
- Useful commands:
 - `systemctl restart apache2`
 - `systemctl stop apache2`
 - `apache2ctl configtest`
- Useful files:
 - `/etc/apache2/sites-available/default-ssl.conf`
 - `/etc/apache2/ssl.crt/`
 - `/etc/apache2/`
 - `/etc/apache2/mods-enabled/ssl.conf`
 - `/var/www/html`

TLS - More advanced ideas

- Discussion TLS man-in-the-middle
- OCSP Stampling

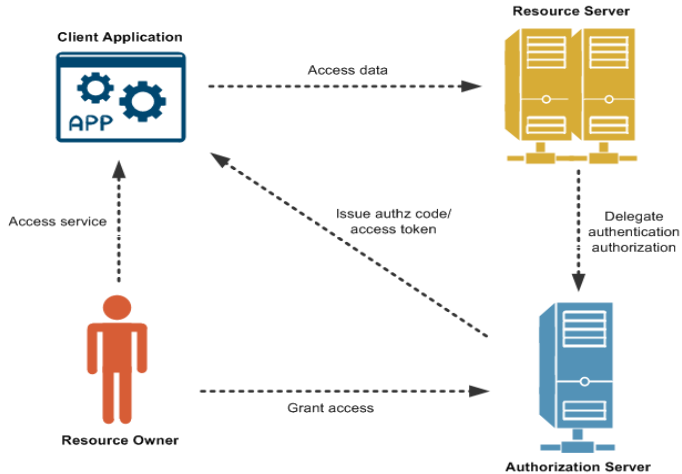
2FA

- Demo Google Authenticator

OAuth 2.0 Agenda

- Demo Adobe EchoSign
- Demo Overleaf
- Discussion about the main advantage
- Keycloak
- Cross-domain authentication system

OAuth 2.0 Overview



TO DO

- Implement an OAuth 2.0 app
- Firstly, use the attached project
- Configure OAuth 2.0 client in Google Cloud Console
- Follow the instructions in readme
- Implement the same flow using other language (.net, java, etc.)